MotionPlay

A Camera-Based Body-Motion Game Controller

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Play Games With Your Body — No Controller Needed

MotionPlay replaces the gamepad with your entire body. An iPhone camera captures your movements, a machine learning model recognizes your gestures in real time, and your actions become live game controller commands — all working with Nintendo Switch emulators.

🎥 Capture

iPhone + ARKit tracks 91 body joints at 60 FPS

🧠 Recognize

ML model classifies gestures in real time

🎮 Control

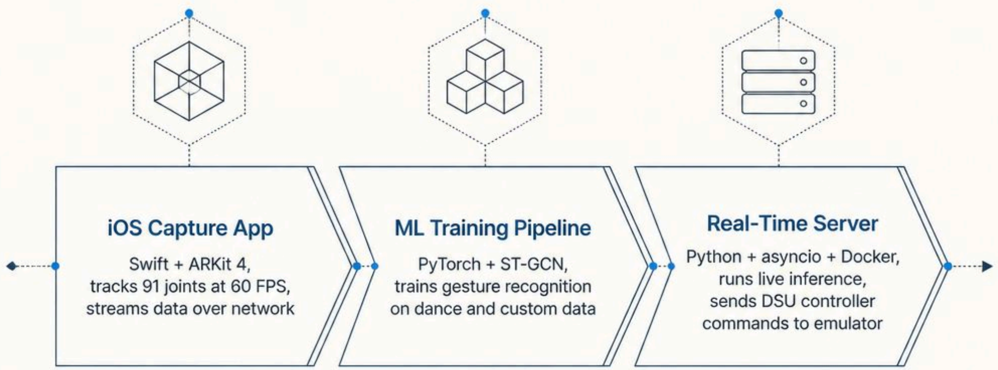
Body movements become game commands

🕹️ Play

Works with Eden / Yuzu Switch emulators

Three Components, One Seamless System

Each piece is independently developed, tested, and deployed — then connected through a clean network protocol.



iOS Capture App

Swift · ARKit 4

ML Training Pipeline

PyTorch · ST-GCN

Real-Time Server

Python · asyncio · Docker

How the Data Flows

- Player Moves
- iPhone ARKit
- Network Stream
- Preprocessing

Total latency: under 100 milliseconds from physical movement to in-game action.

The iOS Capture App

What It Does

- TrueDepth camera tracks a full-body skeleton in real time
- 3D robot character mirrors player movements live
- Records CSV joint data + HEVC video

Network Protocols

- **UDP** — Fastest, no reconnection needed
- **TCP** — Reliable with auto-reconnect
- **WebSocket** — Firewall-friendly, browser support

> Binary frame: 2,652 bytes containing 91 joints (position + rotation) + camera pose. Requires iPhone A12+ / iOS 15+.

ML Training Pipeline: Three Stages

1

Pretraining

ST-GCN trained on **1,408 AIST++ dance clips** across 10 dance styles. Model learns general human movement patterns.

~93% accuracy on dance classification.

2

Template Bank

Custom gesture recordings from 3 subjects. Each recording produces a **256-dimensional embedding**. Embeddings averaged per gesture → template bank.

3

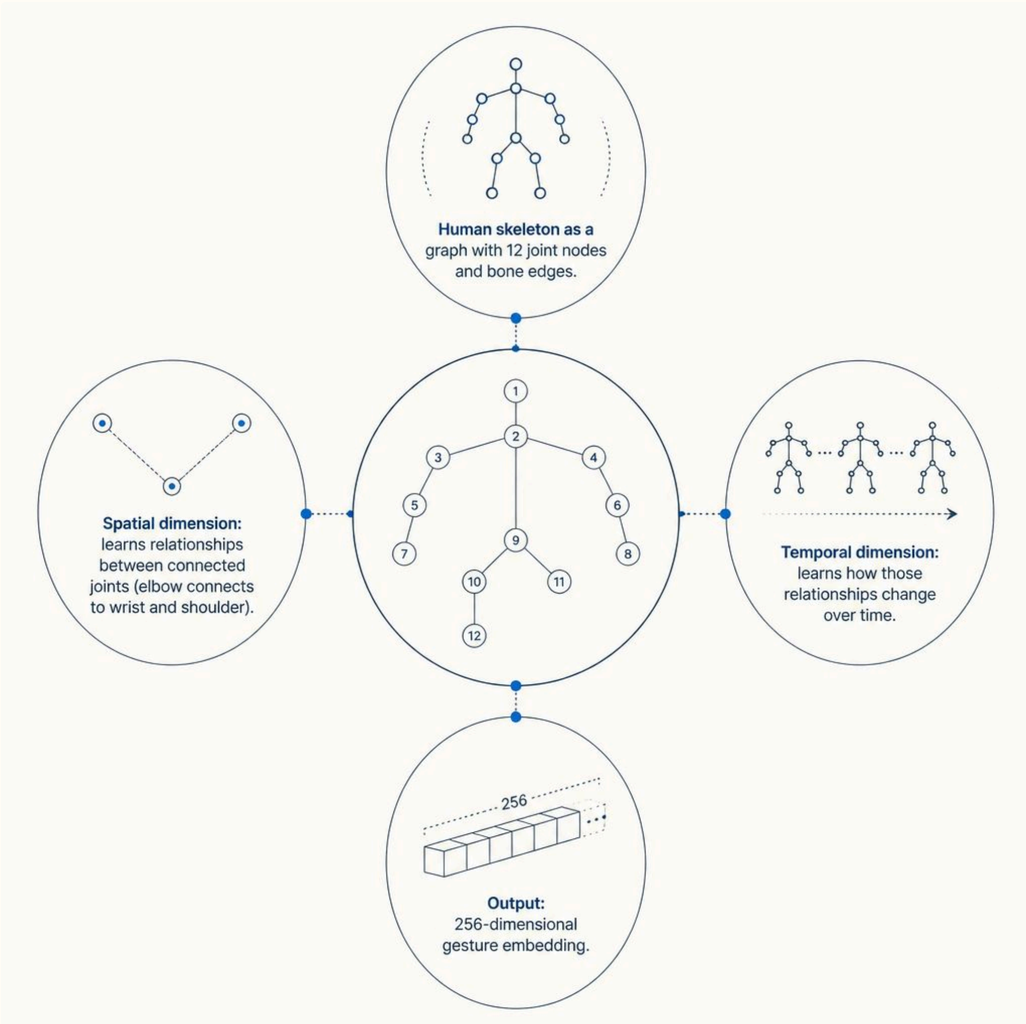
Evaluation

Leave-one-subject-out cross-validation across all 6 gestures.

61% accuracy (6 gestures) · **73% accuracy** (best 4 gestures)

What Is ST-GCN?

Spatial-Temporal Graph Convolutional Network — a neural network designed for skeleton-based action recognition.



Why ST-GCN?

- Works directly on joint coordinates — no video frames needed
- Handles different body sizes through normalization
- Proven on NTU RGB+D and Kinetics-Skeleton benchmarks

Transfer Learning Strategy

Pretrain on large public dataset (AIST++), then fine-tune on our small custom gesture dataset.

The Real-Time Server

01

Receive & Preprocess

Ingests ARKit frames, maps joints, normalizes, and resamples to 300 frames

02

Parallel Recognition

Rule-based tick functions analyze joint angles + accelerations; **ST-GCN** matches embeddings to template bank

03

Score & Select

Gestures compete via **softmax scoring**; winning gesture triggers an action sequence

04

Send to Emulator

DSU protocol at 60 Hz → Nintendo Switch emulator (UDP port 26760)

Bonus: Web dashboard (port 8080) with live 3D skeleton, gesture scores, and virtual gamepad fallback. Fully Dockerized — one command to deploy.

Live Demonstration

What We'll Show

1. **iPhone app running** — 3D robot mirroring player movements
2. **Web dashboard** — Real-time skeleton, gesture scores, button states
3. **Gameplay** — Super Mario 3D World controlled entirely by body

Gesture Mapping

Body Movement	Game Action	Button
Jump up	Mario jumps	A
Quick crouch / squat	Mario dashes	B
Lean / step left	Move left	← Stick
Lean / step right	Move right	→ Stick

Testing Results & What We Found

Test Coverage

223 Unit Tests

DSU protocol + system components

61% / 73%

Cross-user accuracy (6 / 4 gestures)

E2E Tested

Full pipeline with live gameplay

Key Findings

- ✅ **Jump & Dash** — reliably detected via wrist acceleration and hip drop
- ✅ **Left & Right** — detected via arm angle and body lean
- ⚠️ **Forward & Backward** — ARKit reports positions relative to hip, canceling forward movement
- ⚠️ **Jump height invisible** — less than 2 cm vertical hip movement detected

Bugs Fixed

Video recording crash · WebSocket unmasked frames · ST-GCN input shape mismatch · Gesture bank normalization